

Scalability & Future Plan

Date	30 June 2026
Team ID	
Project Name	Smart Lender – AI Powered Loan Eligibility Prediction System
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	SQLite database supports only limited concurrent users.	Performance may decrease as the number of users increases.	High
2	Only the Random Forest model is currently used for prediction.	Prediction accuracy can be improved using advanced machine learning models.	Medium
3	The application is deployed on a free hosting platform with limited resources.	Slower response time during heavy traffic or inactivity.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a small number of concurrent users.	Deploy using cloud infrastructure with load balancing and auto-scaling.
2	Data Storage	Uses SQLite for local data storage.	Migrate to MySQL or PostgreSQL for better scalability and reliability.
3	Performance	Single Flask server handles all requests.	Optimize APIs, use caching (Redis), and deploy with Gunicorn and Nginx.
4	Security	Session-based authentication and password hashing.	Implement JWT authentication, Multi-Factor Authentication (MFA), HTTPS, and role-based access control.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Deploy the application on a production cloud server with a custom domain and HTTPS.	1 Month	Improved availability and accessibility.
Phase 2	Replace SQLite with MySQL/PostgreSQL and optimize database performance.	2 Months	Better scalability, faster queries, and support for more users.
Phase 3	Integrate advanced machine learning models (XGBoost, LightGBM) and add explainable AI features.	3 Months	Higher prediction accuracy and increased transparency in loan decisions.
Phase 4	Develop a mobile application, implement email/SMS notifications, and provide an admin analytics dashboard.	4–6 Months	Enhanced user experience, real-time communication, and better administrative monitoring.